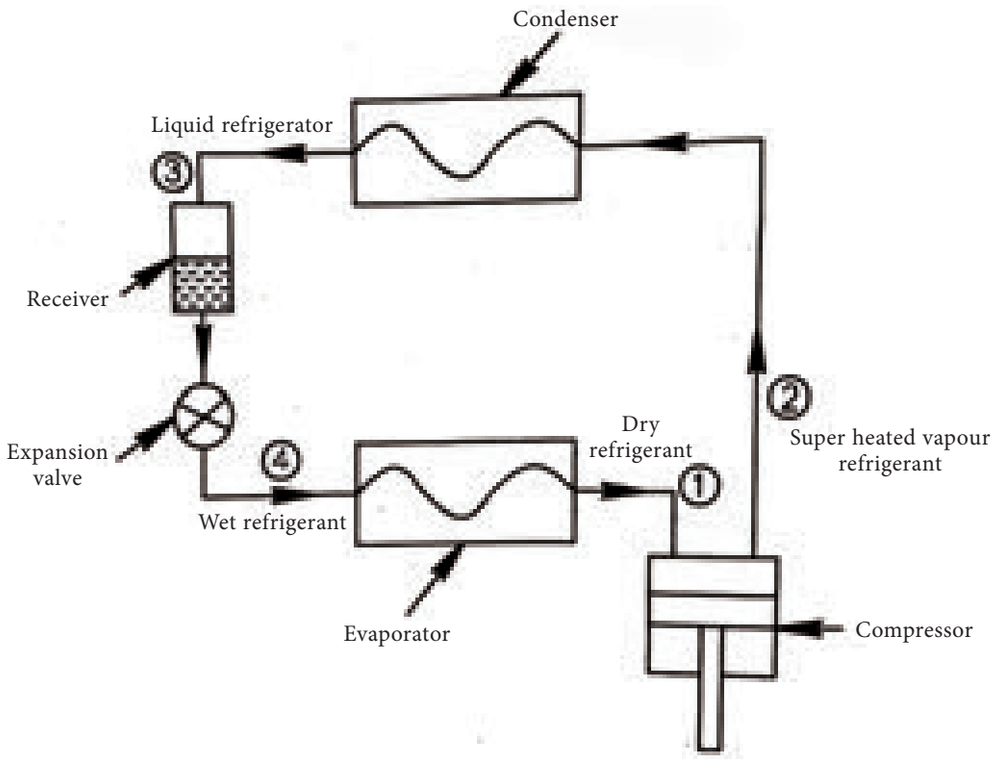

Figure 10.12.1 (b) Simple steam air conditioning system

Important parts of a vehicle airconditioning system

1. Compressor
2. Condenser
3. Receiver/driver
4. Thermostatic Expansion Valve
5. Evaporator
6. Refrigerant
7. Lines and Hoses

The compressor is driven by the engine by means of belt drive. The compressor receives the low pressure, low-temperature vapour refrigerant from the evaporator. This refrigerant is compressed to high pressure, high-temperature vapour refrigerant by the compressor. This refrigerant is sent into the condenser where it is condensed into liquid refrigerant by a constant temperature process.

This liquid refrigerant is expanded in the expansion valve where it is converted into the low-pressure low-temperature liquid refrigerant. This refrigerant is sent into the evaporator, where it receives the heat energy from the passenger compartment and converted into vapour refrigerant, there by cooling effect is obtained. This vapour refrigerant is again sent into the compressor and the cycle is continued. This airconditioning system controls the temperature, humidity, ventilation and dust level simultaneously.

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